

Estimand Divergence: Static Docking Versus Dynamics for Antimalarial Triage

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Abstract

We evaluated a multi-target docking framework in 136 docking systems spanning wild-type and mutant PfDHFR/PfCRT states, built from 396 African natural products and 454 synthetic antimalarials. A Resistance-Resilience Score stratified the primary cohort ($n = 12$) into five tiers that decouple wild-type binding from mutant-state retention. Explicit-solvent MD diverged directionally from static docking in 87.5% (7/8; 95% Wilson CI 52.9–97.8%) of matched states: dynamic pocket relaxation retained poses that static grid scores penalized. Two external checks bounded this signal: DEKOIS 2.0 PfDHFR scoring was at chance level (ROC-AUC 0.450), and rigid $\Delta\Delta G$ did not track published mutant fold-shifts (direction agreement 28.6–41.7%; 24–28 compounds per genotype). At single-replicate scope this divergence is a protocol-

local calibration signal: static scores and short trajectories are complementary, non-equivalent triage metrics.

Introduction

Drug resistance threatens malaria control in Africa, with 282 million cases and 610 000 deaths globally in 2024, concentrated in the WHO African Region¹. Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries². Declining efficacy of some partner drugs has also been reported^{1,3}. Spatial-temporal modelling further indicates that the geographic footprint of *pfk13* markers in Africa is expanding⁴, while genomic surveillance documents the co-occurrence of *Pfprt* and *Pfdhfr* resistance variants across diverse transmission settings⁵⁻⁷.

African natural products represent underexplored antimalarial chemical space, featuring rigid polycyclic frameworks and macrocyclic systems that challenge conventional scoring functions⁸. Multi-target binding strategies aim to raise the mutational barrier to treatment failure⁹, and natural product stereochemical complexity supports activity across diverse protein classes^{9,10}. However, static docking scores capture only rigid pocket complementarity and often fail to account for side-chain reorganization or solvent effects upon mutation^{8,11}. While equilibrium MD filters decoys in cross-docking benchmarks¹², its role in resistance-mutant triage has not been quantified. Recent docking-plus-MD studies of PfCRT and PfdHFR report structural insights without quantifying this concordance^{13,14}.

The upstream library used for this study was generated with generative molecular methods and dual-filter consensus docking (AutoDock Vina¹⁵ and DiffDock-L¹⁶), yielding 65 856 candidate molecules from 396 African natural products and 454 synthetic drugs (upstream library record; see Supporting Information Table S22). We selected the candidate panel against a receptor set spanning established resistance determinants and mechanistically distinct parasite processes: PfdHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT

(chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and PfClpR (PDB 4GM2) represents a proteostasis-related target. This panel provides a mechanistic cross-section to evaluate whether resistance-aware prioritization remains informative across distinct target classes.

This study asks a central question: *do static grid docking and short explicit-solvent molecular dynamics estimate the same target quantity across a resistance-mutant panel?* We test whether their directional disagreement can be used as a calibration flag that prioritizes states for prospective biochemical testing. As a secondary question, we ask whether the multi-metric workflow (docking-derived RRS, PNS, and ACSI) provides discriminant, mutually independent triage dimensions.

We address it in three steps. First, we report *Estimand Divergence*, disagreement between the quantities the two procedures estimate¹⁷: static docking gives a rigid-grid score difference on an unrelaxed receptor, while short MD gives a trajectory-averaged geometric retention ratio. Dynamic pocket accommodation retained poses in 87.5% (7/8; 95% Wilson CI 52.9–97.8%, $n = 8$) of comparable mutant states. Second, we apply the pre-declared WT-eligibility rule, which excludes non-binding wild-type baselines ($|S_{\text{WT}}| < 5.0 \text{ kcal mol}^{-1}$) to prevent artificial score inflation. Third, we test robustness on a 39-compound sensitivity panel (312 docking complexes) rescored by GNINA convolutional neural network (CNN) affinity scoring, with 100% class-level agreement (no per-mutant rank transfer claimed). We treat the docking–MD mismatch not as a computational flaw but as a secondary structural check: short explicit-solvent MD flags static grid artifacts for prospective biochemical testing. At single-replicate scope this mismatch is a calibration signal, not a validated prediction.

Methods

Study design

Our primary analysis focuses on docking-score retention (RRS) for the 12 Set-C candidates with eligible PfDHFR and PfCRT wild-type scores. RRS serves as the baseline for the central comparison, the directional divergence between docking-RRS and short-MD structural retention. We distinguish the primary docking-RRS analysis from the central divergence comparison, secondary structural-retention diagnostics, and technical validation; none of these estimates unobserved biological quantities. The 16-system Set-C MD pilot provides matched comparisons for the divergence signal; the five PfCRT-only candidates and four-complex parent-study MD check are reported separately as secondary analyses. These address different coverage and sampling questions and are not combined with the primary RRS classes. Parent-study MD systems (not among the 17 Set-C candidates) do not validate the Set-C ranking.

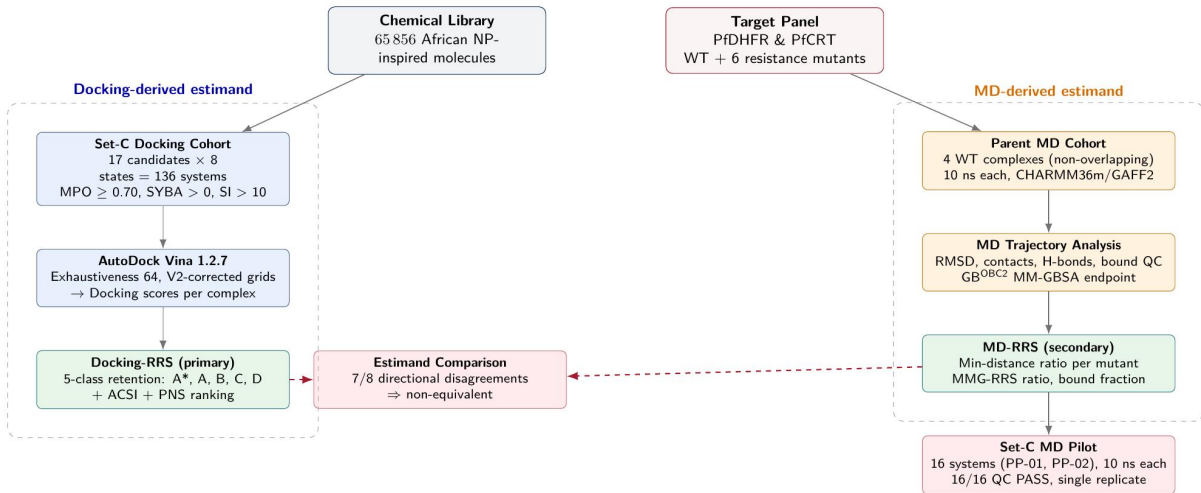


Figure 1: Integrated Resistance-Aware Polypharmacology and Molecular Dynamics Triage Framework. Static multi-target docking (PfDHFR, PfCRT, PfATP4, PfClpR) with Resistance-Resilience Scores (RRS), Antimalarial Chemical Space Index (ACSI), and Polypharmacological Network Score (PNS), followed by 10 ns explicit-solvent MD (OpenFF 2.2.0/CHARMM36m) as a secondary check on the protocol-local Estimand Divergence signal (7/8 matched states). **PP-01** passes the two-target pilot gate; **PP-02** remains single-target scope and was not promoted; **PP-15** is a wild-type feasibility probe outside the Set-C estimand with no RRS estimate. Resistance or evolutionary interpretations require prospective experimental validation.

Library provenance and source-class counts follow the upstream library record (see Supporting Information Table S22). Primary candidates were prioritized using multi-parameter optimization ($\text{MPO} \geq 0.70$, retaining 19 913 primary leads), predicted synthetic accessibility ($\text{SYBA} > 0$), predicted selectivity index ($\text{SI} > 10$), target coverage across at least two docking receptors, and scaffold diversity (≤ 3 compounds per Bemis–Murcko scaffold). This selection yielded the Set-C candidate panel ($n = 17$). Four additional parent-study production-MD wild-type complexes (systems 201, 214, 438, 164; the 438-labelled system contains upstream index 437) were evaluated; these carry no consensus-lead status and their ligand identities and receptor correspondence are documented in the archived ligand identity review. The 164 system represents PfClpR (PDB 4GM2, not PfClpP). Five approved anti-malarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) served as reference controls (Table 1). Class frequencies are conditional on these filters and should not be extrapolated to the full library.

Table 1: Breakdown of the 136 docking protein–ligand systems by target. The 17 Set-C candidates were docked against 5 PfDHFR states (WT + N51I, C59R, S108N, I164L) and 3 PfCRT states (WT + K76T, K76A).

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (Set-C)	17	5	85	—
PfCRT WT + 2 mutants (Set-C)	17	3	51	—
Docking total	—	—	136	—

The 136 docking systems were not subjected to production MD. The 16-system pilot and four non-overlapping parent-study complexes were analyzed separately.

Resistance mutation panel

We selected six mutant or comparator states (Table 2); target identifiers are in the Supporting Information (Table S15). PfDHFR substitutions and PfCRT K76T represent resistance-associated contexts described in WHO resources; K76A is a mechanistic comparator, not a prevalence estimate.

Table 2: Resistance mutation panel.

Target	State	Resistance context	Evidence boundary
PfDHFR	N51I	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	C59R	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	S108N	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	I164L	High-level antifolate-resistance state	Comparator for severe resistance
PfCRT	K76T	Chloroquine-resistance marker	Transporter-resistance context
PfCRT	K76A	Mechanistic comparator for Lys76 perturbation	No clinical prevalence or resistance-marker claim

Homology modelling of resistance mutants

No mutant crystal structures were available for the panel; the six mutant structures were generated with SWISS-MODEL from wild-type templates PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were used as deposited crystal structures with no mutants modelled. PDB 4GM2 is PfClpR, not PfClpP. Models were retained only when QMEAN exceeded -4.0 , GMQE exceeded 0.7 , Ramachandran favored residues exceeded 95% , and backbone RMSD was below $1.5\text{ \AA}$. All six models met these thresholds (Table S3). Model scores are not experimental structures.

Docking protocol assessment

The docking protocol follows the upstream library workflow (see Supporting Information Table S22): AutoDock Vina 1.2.7¹⁵ with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates¹⁸ of PfDHFR (7F3Y), PfCRT (6UKJ¹⁹), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP). The V2 grid specifi-

cation represents the post-review refinement from the companion study, including the critical PfCRT K76→LYS restoration for wild-type reference state and P2Rank-validated binding-site centers. P2Rank²⁰ pocket predictions were compared with the grid centers as a boundary audit of these fixed boxes (Figure S1).

Parent-study enrichment benchmarks (V2 grids) are summarized here and tabulated in full in the Supporting Information (Table S2). On the MMV Malaria Box consensus (Vina+DiffDock), ROC-AUC was 0.924 (PfDHFR), 0.971 (PfCRT), and 1.000 (PfATP4). In contrast, DEKOIS 2.0 enrichment for PfDHFR with Vina alone was null (ROC-AUC 0.450, $EF_{5\%} = 0.00$), indicating that score-based recovery of known actives is protocol- and library-dependent. Redocking was considered accurate when heavy-atom RMSD was $< 2.0 \text{ \AA}$ relative to the crystal pose; the MTX redock attempt was a clear failure (RMSD $\approx 30 \text{ \AA}$). AutoDock Vina was used for all reported Set-C scores; GNINA pose scoring yielded no usable score for the Set-C panel (the secondary 39-compound panel below uses GNINA CNN affinity rescore). Docking scores are empirical scoring approximations that are sensitive to the scoring function and protocol^{8,21,22}; they should not be treated as binding free energies.

Secondary docking calibration

A secondary Tartarus cross-docking analysis used QuickVina on the primary-lead panel. The composite MPO–docking-score association was null in this exploratory check; complete correlations are provided in the Supporting Information (Tables S10 and S11).

Retrospective confrontation with published genotype–phenotype data

As an external, purely retrospective check of the static $\Delta\Delta G$ estimand, we curated IC₅₀ records for *PfDHFR* (ChEMBL target CHEMBL1939²³) by assay description, which encodes the DHFR genotype. The curated set contains 28 compounds with paired wild-type and

quadruple-mutant (N51I/C59R/S108N/I164L) measurements; 24 of them also carry paired double-mutant (C59R/S108N) measurements, and triple-mutant (C59R/S108N/I164L) pairs are complete for all 28. Assay identifiers, raw records, and provenance are frozen with the analysis and described in Data Availability. These are whole-parasite growth IC_{50} values from DHFR-genotyped strains with distinct genetic backgrounds (wild-type TM4/8.2, double-mutant K1CB1, triple-mutant Csl-2, quadruple-mutant Vl/S), not isogenic recombinant-enzyme measurements. We therefore use them descriptively as a phenotype-level reference, not as a binding-affinity ground truth. This genotype-resolved curation differs from the upstream activity-transfer panel audited in the Supporting Information (Table S21), which carries compound-level activity labels without genotype pairing.

The 28 compounds were docked against the wild-type, double, triple, and quadruple *Pf*DHFR states with the declared Vina protocol (AutoDock Vina 1.2.7, exhaustiveness 64, fixed V2 *Pf*DHFR box, single seed 42; 112/112 jobs completed). Double, triple, and quadruple receptors were generated from the wild-type coordinates by in silico side-chain replacement with backbone coordinates unchanged, then prepared identically under the same protocol. For each genotype, $\Delta\Delta G = s_{\text{mutant}} - s_{\text{WT}}$ was compared with the published fold-shift $\log_2(IC_{50,\text{mut}}/IC_{50,\text{WT}})$ using Spearman rank correlation, direction agreement (both quantities positive or both negative), and an exact binomial probability of the observed departure from chance agreement (Table S26).

African Natural Product chemical space and structural descriptors

The candidate panel (Set-C, $n = 17$) was derived from the African Natural Products Database (ANPDB) and synthetic antimalarial reference sets. Physicochemical and stereochemical descriptors were computed using RDKit (v2025.03.6). The cohort exhibits a mean molecular weight (MW) of (271.9 ± 69.3) g/mol, a partition coefficient (LogP) of 2.84 ± 0.82 , a quantitative estimate of drug-likeness (QED) of 0.70 ± 0.06 , a fraction of sp^3 -hybridized carbons (Fsp^3) of 0.22 ± 0.15 , and an average of 0.6 ± 0.9 chiral centers per molecule.

These properties reflect compact, low-molecular-weight natural product cores (including prenylated chromones, flavonoid derivatives, and indole alkaloids) isolated from African medicinal flora such as *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Artemisia* species. The high mean MPO score (0.728 ± 0.021) and synthetic accessibility scores (SYBA = 36.6 ± 23.4) indicate chemical-space alignment with synthetic feasibility.

MPO sensitivity and predicted ADMET profile

MPO-weight sensitivity and single-source ADMET-AI predictions were used descriptively; the complete sensitivity and endpoint results are provided in the Supporting Information (Figure S2 and Table S4). These predictions were not experimentally validated.

Docking-derived resistance-retention score definition

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The 136 systems were ranked with empirical Vina scores rather than calibrated binding free energies. Grid boxes were centered on the co-crystallized ligand position with 25 Å dimensions, matching the audited V2 configuration files¹⁸.

For target t , let $s_{\text{Vina},i,\text{WT},t}$ and $s_{\text{Vina},i,m,t}$ denote the signed Vina scores. Candidates with $|s_{\text{Vina},i,\text{WT},t}| < 5.0 \text{ kcal mol}^{-1}$ are excluded from that target’s score-retention analysis. The target-specific resistance-retention score (RRS) is

$$\text{RRS}_{i,m,t} = \frac{|s_{\text{Vina},i,m,t}|}{|s_{\text{Vina},i,\text{WT},t}|} \times 100. \quad (1)$$

RRS is a relative docking-score-retention statistic, not a free-energy difference or resistance measurement. The available-target summary averages the defined target-specific values across eligible targets and mutant states. The primary analysis is restricted to the 12 candidates with eligible PfDHFR and PfCRT wild-type scores; the five PfCRT-only candidates are reported separately.

The mutually exclusive classes are defined from the available mutant values; primary class counts use the complete two-target subset:

- **Class A***: all available mutant RRS values are at least 80 % and the minimum eligible wild-type score magnitude is at least 7.0 kcal mol⁻¹.
- **Class A**: all available mutant RRS values are at least 80 %, but the minimum eligible wild-type score magnitude is below 7.0 kcal mol⁻¹.
- **Class B**: all available mutant RRS values are at least 70 %, but at least one is below 80 %.
- **Class C**: at least one available mutant RRS value is at least 80 %, but not all available values reach 70 %.
- **Class D**: no available mutant RRS value reaches 80 %.

The classes are mutually exclusive. The cut-offs follow the Vina score distribution in this panel and are not calibrated to biochemical resistance or clinical outcome. Sensitivity to alternative thresholds is in the Supporting Information (Table S6).

Secondary ranking descriptors

ACSI places each candidate relative to approved-drug and African natural-product reference spaces; weight-sensitivity checks are in the Supporting Information (Table S5). PNS is a network-weighted ranking score; PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S9). Associations among metrics were assessed by Spearman rank correlation with 100 000 permutations, Bonferroni adjustment for three tests, and bootstrap intervals from 10 000 resamples. At $n = 12$ these correlations are descriptive and are not powered tests of independence.

MD protocol

We simulated all four parent-study wild-type systems for 10 ns with CHARMM36m²⁴/GAFF2²⁵ at 310.15 K using GROMACS^{26,27}. The 10 ns length is intended as a pose-retention and local-geometry check, not as a measurement of residence time or converged free energy²⁸. We analyzed the trajectories after removing periodic-boundary discontinuities. MDAnalysis²⁹ was used to calculate contacts, H-bonds, RMSF, and radius of gyration every 10 frames and RMSD on every frame.

We obtained MM-GBSA endpoint estimates from production trajectories using the GB^{OBC2} model (igb=5) with 0.15 M salt, via the `gmx_MMPBSA` v1.5.0.3³⁰ CHARMM36-to-AMBER conversion. These are endpoint enthalpic scores, not rigorous free energies, and their limits for ranking dissimilar complexes are well documented³¹. Predicting mutation effects with endpoint methods remains hard, with best-case correlations near 0.44 even at 100 ns³². We did not include configurational entropy. For the interpretable parent-study and PP-15 summaries we report frame-level SEM; the Set-C table instead shows the propagated within-trajectory standard deviation (SD_{prop}). Neither quantity is a replicate-level uncertainty estimate. The inter-replicate diagnostic is reported separately in the Supporting Information.

The 16-system Set-C pilot used the same temperature and length, with ligand parameters from OpenFF 2.2.0 (AM1-BCC) rather than GAFF2. Absolute MM-GBSA values are therefore not compared between the two cohorts. Each mutant was run as a single replicate; trajectories were accepted for endpoint analysis only after geometric quality checks. Trajectory-based retention is summarized as $\text{MD-RRS}_{\text{distance}} = 100 \times \langle d_{\text{min}} \rangle_{\text{mut}} / \langle d_{\text{min}} \rangle_{\text{WT}}$, where d_{min} is the minimum heavy-atom protein–ligand distance; values at or below 100 % indicate pose retention at least as tight as the wild-type endpoint, not increased affinity.

Results

Central Finding: Estimand Divergence Between Static Docking and Dynamic Relaxation

Explicit-solvent molecular dynamics diverged directionally from static grid docking in 87.5 % (7/8; 95% Wilson confidence interval 52.9–97.8 %, $n = 8$) of matched resistance-mutant states. Where static docking predicted reduced score retention (docking-RRS < 100 %, indicating looser mutant binding), short 10 ns MD trajectories retained geometric contact stability (MD-RRS_{distance} $\leq$ 100 %, indicating tighter or equivalent mutant geometry). This directional disagreement reflects distinct computational estimands rigid-grid score differences (ΔG_{grid}) versus trajectory-averaged geometric retention ratios (MD-RRS_{distance}), rather than random error. Dynamic pocket accommodation rescues poses that static scoring over-penalizes, consistent with known docking limitations for stereochemically complex natural products⁸. At single-replicate scope this pattern serves as a protocol-specific calibration signal: the two methods provide complementary triage information for prioritizing resistance-mutant states toward prospective biochemical testing. The divergence interval excludes the 50 % chance level, though it is wide at small sample size. We report this as an observed methodological boundary, not a validated mechanistic prediction (Figure 1; detailed results in the *Estimand Divergence as a Structural Stress-Testing Filter* subsection below).

Operational reproducibility of the docking-RRS signal

Mean target-specific RRS was 75.1 for PfDHFR (12 candidates) and 86.2 for PfCRT (17 candidates); full target-wise summaries are in the Supporting Information (Table S7). The difference reflects docking-score retention between panels, not biological resistance.

On the complete two-target set ($n = 12$) the classes were A*: 1, A: 1, B: 4, C: 5, and D: 1. On all 17 candidates classified on eligible targets, the counts become A*: 5, A: 1, B: 5, C: 5, and D: 1. The five PfCRT-only compounds are not comparable with the two-target

set. Per-compound RRS profiles are shown in Figure S3. Varying the wild-type eligibility threshold from 4.0 kcal mol⁻¹ to 5.0 kcal mol⁻¹ left class counts unchanged. Raising retention thresholds from 70 %/80 % to 80 %/90 % increased Class D from 1 to 9 (Table S6).

The docking-RRS signal is reproducible under the declared protocol: PP-01 multi-seed dispersion is ± 0.05 kcal mol⁻¹ for PfDHFR and ± 0.04 kcal mol⁻¹ for PfCRT, and PP-15 multi-seed dispersion is ± 0.03 kcal mol⁻¹. In a bounded score-perturbation audit, 77.9 % of RRS classes remained unchanged under ± 1 kcal mol⁻¹ perturbation. This characterizes protocol-level consistency, not pose accuracy or binding fidelity.

Chemical-space and network descriptors (secondary)

The mean ACSI for the 17-candidate cohort was 0.543, with two candidates (11.8 %) exceeding 0.70 on the approved-drug/African-NP reference space; weight-sensitivity checks are in the Supporting Information (Table S5). PNS was only weakly associated with RRS on the two-target set ($\rho = -0.2098$, adjusted $p = 1.0000$); PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S9). At $n = 12$ the wide bootstrap intervals are consistent with distinct rather than redundant ranking dimensions, but they are too wide to establish independence.

Cross-metric associations (secondary, post-selection)

These secondary, post-selection comparisons test whether docking-derived RRS aligns with PNS and ACSI within the same cohort; they are not independent validation of any mechanism.

On the two-target set ($n = 12$), all three correlation coefficients were non-significant after Bonferroni correction: PNS-RRS $\rho = -0.2098$, ACSI-RRS $\rho = -0.4056$, RRS-weakest WT score $\rho = -0.1661$ (Table 3). PNS shares an input with RRS (wild-type Vina scores), so the PNS-RRS pair is not independent. On all 17 candidates, PNS-RRS strengthened ($\rho = -0.5588$, adjusted $p = 0.0667$). Unequal target coverage limits interpretation there.

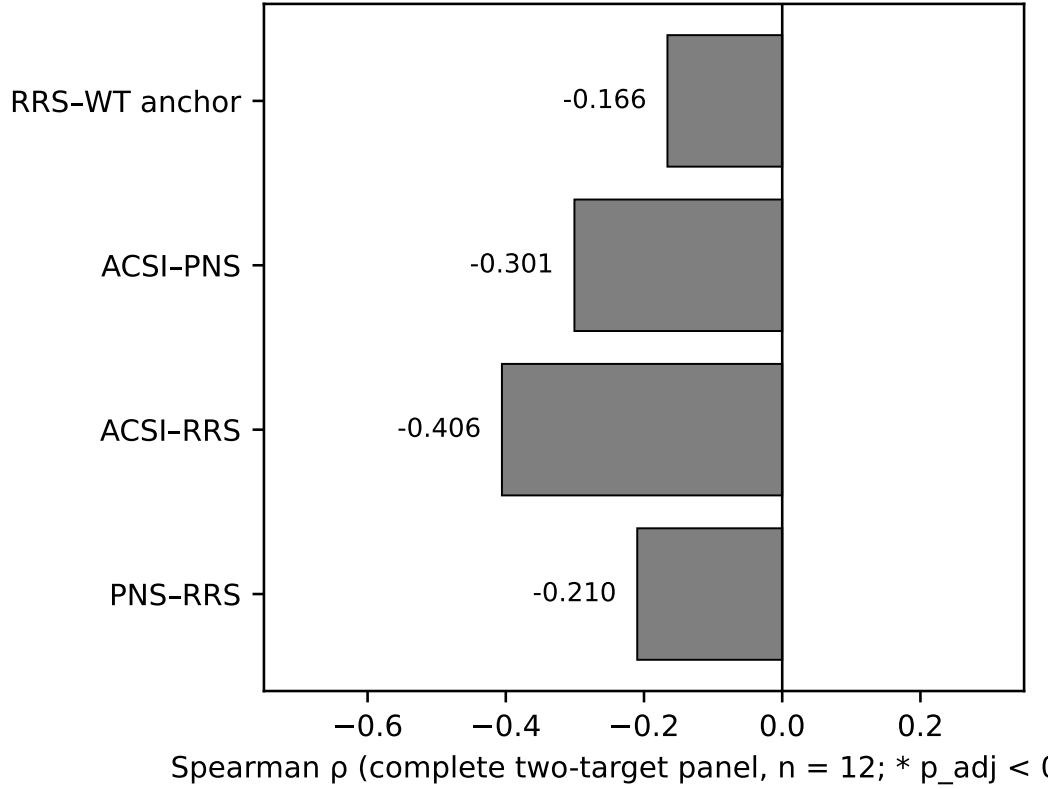


Figure 2: Cross-metric Spearman correlations. Complete two-target panel ($n = 12$): PNS-RRS $\rho = -0.2098$, ACSI-RRS $\rho = -0.4056$, ACSI-PNS $\rho = -0.3007$, RRS-weakest-WT-anchor $\rho = -0.1661$; none significant after Bonferroni correction ($p_{\text{adj}} \geq 0.51$; 100 000 permutations, seed 42). Values from the cross-metric statistical audit.

Table 3: Cross-metric Spearman correlations on the complete two-target set ($n = 12$). Permutation p -values use 100 000 permutations; adjusted p -values use Bonferroni correction for three tests. Bootstrap intervals are percentile 95% intervals from 10 000 resamples.

Metric pair	Spearman ρ	p_{perm}	p_{adj}	Bootstrap 95% CI
PNS vs. RRS	-0.2098	0.5144	1.0000	[-0.76, 0.54]
ACSI vs. RRS	-0.4056	0.1922	0.5766	[-0.89, 0.29]
RRS vs. weakest WT score	-0.1661	0.6038	1.0000	[-0.73, 0.59]

Computational prioritization intersection (exploratory)

Requiring Class A or A*, ACSI > 0.5, and above-mean PNS leaves one candidate on the two-target panel. That intersection is reported as a shortlist for experiment, not as evidence of activity or resistance resilience.

Parent-study MD check (non-overlapping cohort, secondary)

Four non-overlapping parent-study wild-type complexes were simulated for 10 ns as a secondary pose-retention check; these compounds are not among the 17 Set-C candidates and do not validate the Set-C ranking.

Table 4: Production MD stability metrics for parent-study wild-type complexes (10 ns). BB RMSD, ligand RMSD, minimum distance, contacts, and H-bonds are computed from PBC-unwrapped trajectories.

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (system 164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (system 201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (system 214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (system 438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

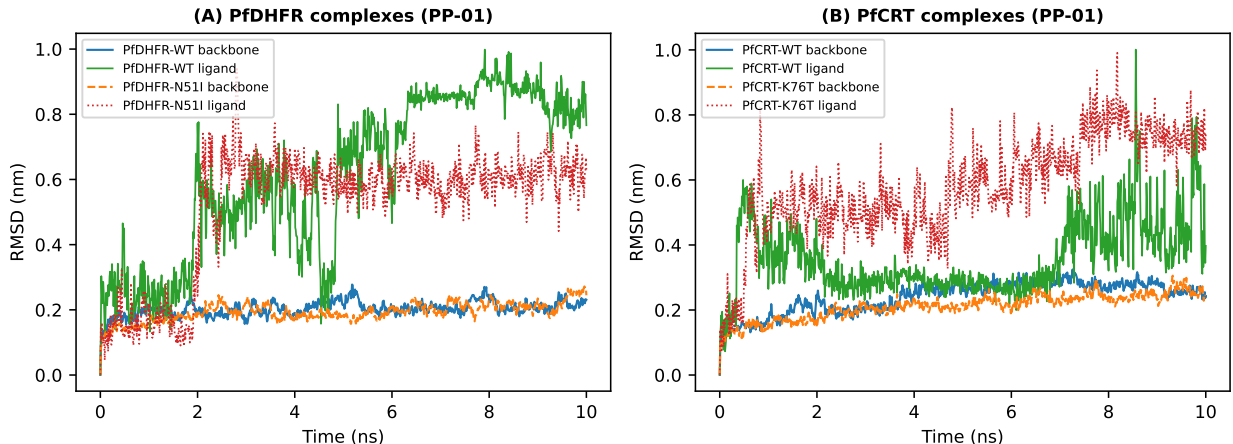


Figure 3: Pose retention and thermal equilibration (canonical R1 trajectories). Explicit-solvent backbone (solid) and ligand-after-backbone-fit (dashed/dotted) RMSD time-series (10 ns) for PP-01, computed with *gm̃ rms* (PBC-whole reconstruction; fit group = Backbone; RMS groups = Backbone / MOL0). (A) PfDHFR: backbone (0.203 ± 0.024) nm (WT), (0.193 ± 0.029) nm (N51I); ligand (0.613 ± 0.240) nm and (0.519 ± 0.190) nm. (B) PfCRT: backbone (0.241 ± 0.048) nm (WT), (0.211 ± 0.039) nm (K76T); ligand (0.357 ± 0.113) nm and (0.581 ± 0.155) nm. All trajectories pass the geometric QC gate (mean minimum heavy-atom distance 2.7–3.2 Å, bound fraction 1.000); ligand RMSD reflects in-pocket mobility, not dissociation.

Table 5: Endpoint energy evaluation for four reference wild-type complexes. Only the PfCRT complex yields an interpretable MM-GBSA endpoint; dissociated systems and conversion anomalies are N/A.

System	Target	ΔG_{bind} (kcal mol ⁻¹)	SEM	Interpretation
System 214	PfCRT-WT	-18.25	0.40	Stable anchorage; ligand bound [†]
System 164	PfClpR-WT	–	–	N/A; ligand dissociated [‡]
System 201	PfDHFR-WT	–	–	N/A; ligand dissociated [§]
System 438	PfATP4-WT	–	–	N/A; parameter conversion anomaly*

[†]Ligand remains bound (minimum distance 3.19 Å, 78 contacts); calculated via `gmx_MMPBSA`. Uncertainty is the SEM over 101 stored snapshots.

[‡]Ligand separates beyond 60 Å minimum distance with zero contact persistence; excluded from binding energy evaluation.

[§]As [‡] (PfDHFR system).

*Excluded: multi-chain membrane-transporter parameterization introduced force-field conversion artifacts.

Of the four reference complexes, two retained stable protein–ligand contacts while two dissociated (Table 4); for PfATP4, force-field parameter conversion precluded reliable MM-GBSA endpoints (Table 5).

Estimand Divergence Between Static Docking and Dynamic Trajectories

The primary docking retention metrics and trajectory-based geometric stability summaries are non-equivalent computational estimands. Explicit-solvent MD was run on PP-01 and PP-02 across 16 receptor states; all 16 trajectories satisfied geometric contact stability thresholds (<3.5 Å minimum protein–ligand distance).

For the eight mutant states with an eligible wild-type docking baseline, static grid docking predicted reduced score retention ($RRS < 100\%$). Explicit-solvent relaxation pointed the other way in 87.5% (7/8; 95% Wilson CI 52.9–97.8%, $n = 8$) of matched comparisons ($MD\text{-}RRS_{\text{distance}} \leq 100\%$). The interval excludes the 50% chance level, though it is wide at $n = 8$. At single-replicate scope this mismatch is an observed calibration signal, not a demonstrated mechanism. Static docking scores (ΔG_{grid}) and trajectory-derived retention ratios ($MD\text{-}RRS_{\text{distance}}$) are complementary triage signals, not interchangeable measures of

binding free energy.

We ran a single-ligand feasibility probe (**PP-15**, outside the 16-system Set-C MD array and the Set-C RRS estimand) on wild-type *Pf*DHFR (7F3Y) and *Pf*CRT (6UKJ). Both systems completed 10 ns production trajectories with 100 % QC-PASS status (bound fraction = 1.000). Mean minimum heavy-atom distances were 2.77 Å (*Pf*DHFR) and 3.20 Å (*Pf*CRT); MM-GBSA endpoints were (-29.52 ± 0.33) kcal mol⁻¹ (*Pf*DHFR) and (-27.79 ± 0.49) kcal mol⁻¹ (*Pf*CRT; Table S20). Multi-seed redocking confirmed docking score stability (± 0.03 kcal mol⁻¹; Table S23). We report no mutant or RRS estimate for this probe and make no dual-target resilience claim from it.

Table 6: Set-C pilot endpoint summaries (exploratory; not used to assign RRS classes). ΔG_{bind} is the mean over 100 snapshots $\pm SD_{\text{prop}}$ (within-trajectory SD, not SEM or replicate-level uncertainty). The PP-01 *Pf*CRT K76A endpoint $(-35.29$ kcal mol⁻¹) carries a 7.75 kcal mol⁻¹ inter-replicate sensitivity (second trajectory -27.54 kcal mol⁻¹; Table S19) and supports no resilience claim. MMG-RRS is the mutant/wild-type endpoint ratio ($\times 100$; wild type = 100). Docking RRS is shown for comparison (n/a: no wild-type docking reference for PP-02 *Pf*DHFR). Single-replicate summaries are within-protocol contrasts only.

Candidate	Target	State	ΔG_{bind} (kcal mol ⁻¹)	SD_{prop}	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	—
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	—
PP-02	PfCRT	K76A	-24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	—
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	-30.16	2.06	100.0	—

Additional docking sensitivity panel

The 39-ligand docking sensitivity panel produced 312 finite docking scores across the same eight *Pf*DHFR/*Pf*CRT states. Bootstrap mean RRS was 100.45 (99.59–101.32) and the Class-A fraction was 0.974 (0.923–1.000). A mean near 100 means that mutant and wild-type Vina scores were on average nearly proportional under this protocol. This is a compu-

tational sensitivity check, not validation of the Set-C ranking or of experimental resistance (Table S21).

Retrospective comparison against published *Pf*DHFR fold-shifts

The curated ChEMBL panel asked directly whether static $\Delta\Delta G$ reproduces published genotype-level resistance shifts (Table S26). It did not. Spearman ρ between $\Delta\Delta G$ and the $\log_2$ fold-shift was 0.19 ($p = 0.375$, $n = 24$) for the double mutant, -0.04 ($p = 0.853$, $n = 28$) for the triple mutant, and -0.13 ($p = 0.504$, $n = 28$) for the quadruple mutant. Direction agreement was 10/24 (41.7%), 9/28 (32.1%), and 8/28 (28.6%), respectively; for the triple and quadruple mutants the rate departs from chance toward disagreement (exact binomial $p = 0.044$ and $p = 0.018$; double $p = 0.271$). These three per-genotype tests are unadjusted: Bonferroni correction gives 0.132 and 0.054 for the triple and quadruple mutants, so we report the rates as descriptive evidence of below-chance agreement rather than formal rejections. Reference DHFR inhibitors pyrimethamine (919-fold) and cycloguanil (145-fold) were mispredicted directionally at every genotype. Mutant scores averaged slightly more favorable than wild-type scores (mean $\Delta\Delta G$ -0.19 – -0.12 kcal mol $^{-1}$ across genotypes; 68–79% of compounds below zero), a systematic shift smaller than the ± 1 kcal mol $^{-1}$ score sensitivity of the protocol. Given the phenotype-level, non-isogenic reference, this comparison stays descriptive: it tests the static $\Delta\Delta G$ estimand against published measurements, not the Set-C candidates.

Discussion

Resistance-Resilience Stratification and Protocol Robustness

The Resistance-Resilience Score (RRS) stratifies the primary candidate cohort ($n = 12$) into five tiers (A*–D) decoupling wild-type binding strength from mutant-state score retention. Under the declared protocol, multi-seed dispersion remained within ± 0.05 kcal mol $^{-1}$ (PP-01

and PP-15 across PfDHFR and PfCRT), and 77.9 % of class assignments were invariant to $\pm 1.0 \text{ kcal mol}^{-1}$ score perturbations.

Protocol robustness was tested on an external 39-compound sensitivity panel (312 docking complexes; Table S21). GNINA CNN cross-scoring showed 100 % class-level agreement with Vina, with no per-mutant rank transfer claimed (per-mutant ρ near zero for C59R/I164L). Because almost the whole panel falls in Class A (bootstrap Class-A fraction 0.974), this agreement is close to degenerate and constrains only gross class assignment, not ranking quality. On the DEKOIS 2.0 PfDHFR benchmark, Vina ROC-AUC was 0.450 ($\text{EF}_{5\%} = 0.00$) chance level, cautioning against treating RRS as a calibrated predictor outside the primary cohort; MD is reported as a parallel structural check, not a validation of docking classes.

Our candidates originate from African natural-product chemical space, without claiming ethnobotanical validation. The panel’s drug-likeness ($\text{QED} = 0.70 \pm 0.06$) and synthetic accessibility ($\text{SYBA} = 36.6 \pm 23.4$) are consistent with actionable synthetic feasibility; confirmation requires prospective experiment.

A retrospective confrontation with published genotype-resolved *PfDHFR* fold-shifts (ChEMBL, 28 paired compounds) reached the same conclusion from experimental data: static $\Delta\Delta G$ did not track whole-parasite resistance shifts at any genotype ($\rho -0.13$ – 0.19 , direction agreement 28.6–41.7 %). This matches the chance-level DEKOIS enrichment above and the estimand-divergence framing: a single rigid-grid score difference does not estimate phenotype-level resistance. The comparison neither validates nor invalidates the RRS tiers of the primary cohort. RRS is a within-protocol score-retention statistic on anchored poses for the disjoint Set-C compounds; the ChEMBL endpoint is whole-parasite growth in non-isogenic backgrounds, where permeability, efflux, and background mutations contribute. The two analyses address different estimands (Table S26). We performed no experimental measurement on the Set-C candidates themselves; prospective validation remains the outstanding anchor (see Limitations).

Estimand Divergence as a Structural Stress-Testing Filter

We report *Estimand Divergence* between static docking RRS and short explicit-solvent MD pose retention ($\text{MD-RRS}_{\text{distance}}$) as a protocol-local observation. In 87.5 % (7/8; 95% Wilson CI 52.9–97.8 %, $n = 8$) of matched mutant states, grid docking predicted score penalties ($\text{RRS} < 100$ %) while MD retained poses ($\text{MD-RRS}_{\text{distance}} \leq 100$ %); the Wilson interval excludes the 50 % chance level.

Rigid grid docking evaluates unrelaxed steric complementarity (ΔG_{grid}), whereas short MD captures local conformational adaptation, hydrogen-bond rearrangement, and solvent-shielded stabilization. Per the JCIM MD reporting guidelines¹¹, the 10 ns trajectories are secondary structural checks, consistent with established MD-based pose-stability filtering of docked poses¹² but applied here to resistance-mutant triage rather than pose selection. A thermal noise floor of $2.01 \text{ kcal mol}^{-1}$ from independent WT replicates (PP-01 PfCRT WT R1/R2) bounds numerical variation of end-state estimates at this scope. Contemporaneous docking-plus-MD surveys of resistance mutants report qualitative structural conclusions without quantifying this concordance^{13,14}.

This pattern is consistent with docking limitations for stereochemically complex natural products⁸: scoring functions calibrated on synthetic drug-like molecules can over-penalize rigid polycyclic frameworks. At single-replicate scope this is a within-protocol contrast, not a general docking-failure rate.

The Class-A* lead **PP-01** illustrates this across its eight-system panel (Table S25). Static docking penalized *PfDHFR* C59R ($\text{RRS} = 85.5$ %) and *PfCRT* K76T ($\text{RRS} = 92.5$ %), yet 10 ns trajectories retained both poses (100 % QC-PASS, bound fraction 1.000). Mean minimum heavy-atom distances were 2.91 Å (C59R; $\text{MD-RRS}_d = 98.9$ %) and 3.02 Å (K76T; $\text{MD-RRS}_d = 102.8$ %), with MM-GBSA endpoints (-26.56 ± 1.91) kcal mol^{-1} (C59R) and (-29.51 ± 2.65) kcal mol^{-1} (K76T). This shows geometric pose retention, not thermodynamic proof: the K76A endpoint carries a $7.75 \text{ kcal mol}^{-1}$ inter-replicate sensitivity, so we make no MM-GBSA-derived resilience claim for K76A. Biological resistance conclusions require

experimental validation.

ProLIF interaction fingerprints across all 16 QC-PASS trajectories (100 snapshots per system) show a sparse conserved contact network (Figure 4). *PfCRT* Tyr16 maintained 100 % contact occupancy in five of six *PfCRT* systems (PP-01/PP-02 $\times$ WT/K76T/K76A); *PfDHFR* Asp54, Leu46, Met55, and Ile14 reached full occupancy across the PP-01/PP-02 states. This contact pattern is compatible with pose retention but does not establish a general mechanism.

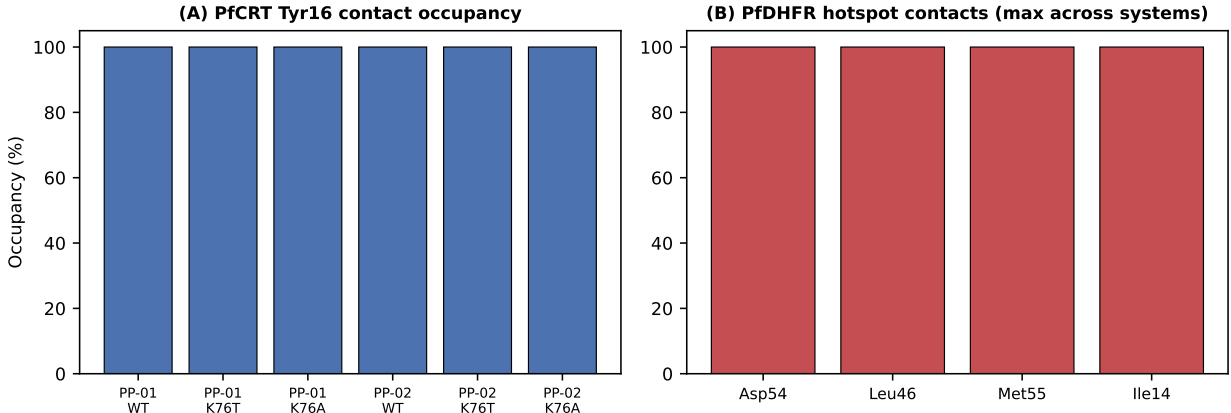


Figure 4: Conserved hotspot anchors. *ProLIF* interaction-fingerprint occupancy (100 snapshots/system; hydrophobic, van der Waals, and hydrogen-bond contact types). (A) *PfCRT* Tyr16 contact occupancy per system: 100 % in five of six *PfCRT* systems. (B) *PfDHFR* hotspot contacts (Asp54, Leu46, Met55, Ile14): maximum occupancy across the PP-01/PP-02 states. Source: *ProLIF* interaction-fingerprint analysis of the QC-PASS trajectories.

Orthogonal Multi-Dimensional Triage

RRS, ACSI, and PNS show non-significant pairwise associations on the primary cohort ($n = 12$; $\rho \in [-0.41, -0.21]$, $p_{\text{adj}} > 0.50$; Table 3), consistent with distinct ranking dimensions. In an exploratory post hoc check, controlling for molecular weight and scaffold frequency gave a moderate partial PNS–RRS association ($\rho_{\text{partial}} = -0.6154$, descriptive only, no formal test); it does not overturn the pre-declared null result above.

STRING threshold sensitivity (confidence cutoffs 400, 700, 900) showed stable PNS ranks (ρ 0.9632–0.9975; 400-vs-700 0.9975, 700-vs-900 0.9681, 400-vs-900 0.9632). Threshold choice

therefore does not alter ranking within this panel. This differs from the PfCRT-centrality imputation sensitivity (Table S9).

Explicit-solvent MD evaluates dynamic stability without reliance on training distributions, providing a complementary check on static scores.

Mechanistic Context and Multi-Target Positioning

The retained PfCRT poses are consistent with microsecond-scale transporter studies³³: Lys76 anchors the central cavity, and K76T/K76A alter local charge density and hydration, permitting ligand reorientation that preserves hydrophobic contacts despite grid-level penalties.

Within a single-drug-multi-target framing⁹, Class-A* candidate PP-01 is prioritized for assay follow-up: it retained multi-target binding potential across mutant landscapes in these simulations. Whether dual-target engagement raises any evolutionary barrier to resistance cannot be determined computationally. It requires prospective biochemical, structural, and *in vitro* evaluation. We make no resistance, fitness, or evolutionary claim; PP-15, a wild-type probe outside the Set-C estimand, carries no such interpretation.

Practical Guidance

The combined results translate into four operational rules for resistance-aware triage pipelines. (i) Do not rank resistance variants by rigid-receptor $\Delta\Delta G$ alone: it failed both an external enrichment benchmark (DEKOIS 2.0 PfDHFR ROC-AUC 0.450) and a retrospective phenotype comparison (ρ -0.13 – 0.19). (ii) Treat disagreement between static docking and short explicit-solvent MD as a flag for experimental follow-up rather than as evidence against either method, consistent with established MD-based pose filtering¹². (iii) Report RRS, PNS, and ACSI as separate descriptive dimensions rather than combining them into a single index, given their non-significant mutual associations at cohort scale. (iv) Enforce the WT-eligibility gate ($|S_{WT}| \geq 5.0 \text{ kcal mol}^{-1}$) before computing any ratio-based score, to

prevent inflation on weak wild-type baselines. Table 7 states, for each central claim of this study, the evidence and its status under the analyses reported here.

Table 7: Claim ledger. *Evidential status of the central claims under the analyses reported in this study. Statuses are scope-qualified: “supported” means supported within the declared protocol and sampling limits, not validated prediction.*

Claim	Evidence reported here	Status
Static docking and short MD estimate the same target quantity	7/8 matched mutant states diverge; 95% Wilson CI 52.9–97.8% excludes 50% chance ($n = 8$)	Refuted at this scope
Vina enriches near-native poses on DEKOIS 2.0 PfDHFR	ROC-AUC 0.450; EF _{5%} = 0.00 (chance level)	Refuted
Rigid-receptor $\Delta\Delta G$ tracks resistance fold-shifts	ChEMBL: ρ -0.13 – -0.19 ; direction agreement 28.6–41.7%	Refuted
RRS, PNS, and ACSI are mutually independent discriminant dimensions	$\rho \in [-0.41, -0.21]$, $p_{\text{adj}} > 0.50$ ($n = 12$)	Not supported
Class assignments survive independent rescoring	GNINA 100% agreement but Class-A fraction 0.974	Supported, near-degenerate
PP-01 retains poses across matched mutant states	8/8 QC-PASS (6 matched mutant states); bound fraction 1.000 in 10 ns trajectories	Supported (protocol-local)
MM-GBSA endpoint resilience (K76T, C59R)	Within the 2.01 kcal mol ⁻¹ WT noise floor; K76A sensitivity 7.75 kcal mol ⁻¹	Supported at pilot scope; no claim for K76A
Multi-target engagement raises an evolutionary barrier to resistance	Not determinable computationally	Open (prospective)

Limitations

1. **Receptor models:** mutant structures are homology models (quality criteria in Methods), not experimental structures.
2. **Timescale and replication:** per JCIM guidelines¹¹, the 10 ns single-replicate pilot is a structural stress test, not a converged free energy or residence-time measurement. The PP-01 PfCRT K76A endpoint (-35.29 kcal mol⁻¹) differs by 7.75 kcal mol⁻¹ from a second trajectory (-27.54 kcal mol⁻¹; Table S19), which exceeds the 2.01 kcal mol⁻¹ WT noise floor. We therefore make no endpoint resilience claim for K76A.
3. **Sample size:** primary two-target cohort $n = 12$, plus 5 single-target candidates and a 39-ligand external panel.
4. **Protonation:** preparations at physiological pH; PfCRT dynamics in the acidic digestive vacuole (pH 5.2) may differ.

5. **Retrospective experimental reference:** the ChEMBL comparison (Table S26) rests on whole-parasite IC₅₀ values from non-isogenic DHFR-genotyped strains; it is a descriptive phenotype-level reference, not enzyme-level or cohort-specific validation. No experimental measurement was performed on the Set-C candidates, which therefore remain without their own experimental anchor. ChEMBL docking used a single seed (42).

Conclusion

This study reports a resistance-aware computational triage workflow for African antimalarial natural products. Static grid docking (ΔG_{grid}) and short explicit-solvent MD pose retention (MD-RRS_{distance}) capture distinct, non-equivalent signals: dynamic pocket accommodation retained poses where static docking predicted penalties in 7/8 matched mutant states (95% Wilson CI 52.9–97.8%, $n = 8$). The multi-metric descriptors stayed mutually non-associated at cohort scale ($\rho \in [-0.41, -0.21]$, adjusted $p > 0.50$), so we report them as distinct descriptive dimensions rather than a combined index. **PP-01** (Class A*) showed multi-target engagement across PfDHFR and PfCRT mutant panels and passes the two-target pilot gate (Class-A* retention plus pose retention in matched mutant trajectories). Whether this implies any elevated evolutionary barrier requires prospective biochemical, structural, and *in vitro* evaluation; these findings prioritize specific natural product scaffolds for that testing. External checks bounded the docking estimand at chance level (DEKOIS 2.0 PfDHFR ROC-AUC 0.450; ChEMBL direction agreement 28.6–41.7%), and no experimental measurement was performed on the Set-C candidates themselves.

Authors’ Contributions

Myke Vital Sao Temgoua: Methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapel Njafa: Methodology, software, super-

vision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Fon Wilfred Mbacham: conceptualization, validation, writing – review & editing. Serge Guy Nana Engo: conceptualization, methodology, supervision, writing – review & editing. All authors read and approved the final manuscript.

Data Availability

Docking results, metric outputs, analysis scripts, and source-data records are publicly archived on Zenodo (dataset CC-BY-4.0; analysis code MIT) at DOI: [10.5281/zenodo.22829031](https://doi.org/10.5281/zenodo.22829031). Large parent-study trajectories are excluded; conclusions are restricted to the trajectory-derived summaries reported above. The retrospective ChEMBL comparison (curated paired IC₅₀ inputs with assay provenance, 112 docking scores, analysis outputs, and a script and parameter hash manifest) is archived with the source-data records. The parent-study ligand identity review (production-ligand composition, numbering provenance, and receptor-correspondence checks) is archived with the same records.

Competing Interests

The authors declare no competing interests.

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During the preparation of this manuscript in September 2026, the authors utilized AI tools to support code review, provide structured text suggestions, and perform consistency checks. The authors designed all analyses, maintained full control over the scientific inter-

pretation, and rigorously verified all computational results and quantitative claims against the original source records.

Supporting Information Available

Docking enrichment and homology-model quality tables; P2Rank box audit; MPO, ADMET, and ACSI sensitivity; RRS radar profiles and threshold sensitivity; PNS imputation and Tartarus calibration; cohort and evidence-scope tables; PlasmoDB identifiers; parent-study and Set-C pilot MD/MM-GBSA summaries; robustness and upstream-workflow-context tables; retrospective ChEMBL genotype–phenotype fold-shift table (PDF).

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TOC Graphic

